Unit: 6 Unit 8: Quadratics **Name:** ________________
Topic: Converting Forms (Completing the Square) **Date:** ____________

Foundation (Jalapeno)

Q1. Convert $x^2 + 6x + 8$ to vertex form.
(A) $(x + 3)^2 - 1$
(B) $(x + 3)^2 + 1$
(C) $(x + 4)^2 - 8$
(D) $(x + 4)^2 + 8$
(E) $(x + 2)^2 - 4$

Q2. Complete the square for $x^2 + 4x$.
(A) $(x + 2)^2 - 4$
(B) $(x + 2)^2 + 4$
(C) $(x + 1)^2 - 1$
(D) $(x + 1)^2 + 1$
(E) $(x + 3)^2 - 9$

Q3. What is the vertex of the parabola $y = (x + 2)^2 + 1$?
(A) $(-2, 0)$
(B) $(-2, 1)$
(C) $(-1, 0)$
(D) $(-1, 1)$
(E) $(-3, 0)$

Q4. Convert $y = x^2 + 2x - 6$ to vertex form.
(A) $(x + 1)^2 - 7$
(B) $(x + 1)^2 + 7$
(C) $(x - 1)^2 - 7$
(D) $(x - 1)^2 + 7$
(E) $(x + 2)^2 - 10$

Q5. Complete the square for $x^2 - 8x$.
(A) $(x - 4)^2 - 16$
(B) $(x - 4)^2 + 16$
(C) $(x - 2)^2 - 4$
(D) $(x - 2)^2 + 4$
(E) $(x - 3)^2 - 9$

Q6. What is the equation of the axis of symmetry for the parabola $y = (x - 3)^2 + 2$?
(A) $x = -3$
(B) $x = -2$
(C) $x = -1$
(D) $x = 1$
(E) $x = 3$

Q7. Convert $y = x^2 - 4x - 3$ to vertex form.
(A) $(x - 2)^2 - 7$
(B) $(x - 2)^2 + 7$
(C) $(x + 2)^2 - 7$
(D) $(x + 2)^2 + 7$
(E) $(x - 1)^2 - 4$

Q8. What is the vertex of the parabola $y = (x - 1)^2 - 3$?
(A) $(-1, -3)$
(B) $(-1, 3)$
(C) $(1, -3)$
(D) $(1, 3)$
(E) $(0, -3)$

Open Ended Questions (FRQ)

Q9. Convert $y = x^2 + 10x + 24$ to vertex form and identify the vertex.

Q10. Complete the square for $x^2 - 6x$ and write the resulting equation in vertex form.

Chef's Tips

- Tip 1: Remind students to show all work and use the correct notation.
- Tip 2: Encourage students to check their work by plugging in values.
- Tip 3: Provide feedback on the importance of accuracy and neatness.